Jialin Song
COSI 295A Computational Linguistics Capstone Project

Capstone Project Report: Language Model's Ability on Interpreting Lexicalized Noun-Noun Compounds

1. Introduction

Comprehension of noun-noun compounds (e.g., *birthday cake*) has been a hot topic in psycholinguistics. These compounds generally contain two components – head and modifier. For example, in the compound of *birthday cake*, *birthday* is the modifier, and *cake* is the head. Humans interpret compounds differently for different combinations of nouns. Specifically, when a modifier-head compound appears, humans are always able to assign an implicit relation between them. For example, *birthday cake* is "a cake for *birthdays*", but *carrot cake* is "a cake *made with* carrots" instead of "a cake *for* carrots" (Rambelli et al., 2024). This interpretation not only appears in the compounds that speakers are familiar with (i.e., lexicalized), but also appears in those that sounds less sensible to speakers (i.e., unlexicalized or novel) – *mountain magazines* is likely to be interpreted as "magazines *about* mountains," even if this compound is uncommonly seen in our daily life (Gagné and Shoben, 1997).

As the transformer language models achieve more human-expected results, some researchers are interested in whether these models can capture the implicit relations in lexicalized noun-noun compounds. Among these studies, one of them is Rambelli et al. (2024), which tested on some trending large language models (LLMs), specifically Llama-2, Mistral, and Falcon, and used BERT-large and GPT-2 as baselines. They collected lexicalized English compounds with human-annotated implicit relation from previous published datasets, reannotated them with another set of cross-linguistically consistent relations, and gave each compound a correct paraphrase and a list of incorrect paraphrases. They evaluated the performance in two ways. First, they extracted on the surprisal scores, a metric based on the log probabilities of LLMs, and see if models can differentiate the correct paraphrases of a compound from incorrect ones. Resulting accuracy scores showed that LLMs all performed significantly better than baselines, with LLMs' performance ranged from 0.38 to 0.45 while baselines only achieved 0.26 and 0.33. Second, they prompted the instruct-variant of each LLM (i.e., the variant of a certain LLM designed to do text generation or question answering) to choose the correct paraphrases among a mix of correct and incorrect ones, and measured how many correct choices were made out of the total choices. Results with this method were inconsistent: Mistral model could reach the accuracy of 0.80, while Falcon model could only reach 0.15.

Inspired by Rambelli et al. (2024), this capstone intends to experiment with similar methods but different models to answer some other questions that are also related to the interpretation of implicit relations in lexicalized noun-noun compounds. Specifically:

i. Can attention-free models perform as well as transformer models in implicit relation prediction?
ii. Can bilingual or mulitlingual attention models predict as well as monolingual equivalent models in all languages in implicit relation prediction?

The following content is structured as follows: Section 2 introduces more details on the English dataset and Section 3 on the surprisal scores, as they were used in both experiments. Section 4 and 5 answers the research questions above with a review of experiments, data analysis, and results discussion. Section 6 concludes.

2. Dataset from Rambelli et al. (2024)

Scholars have proposed roughly two different ways of interpretation for implicit relations. One of them is ontological, where people list an inventory of relations and assign each compound a correct relation. Problem with this method is that it only captures part of the semantic meaning. For example, in Gagné and Shoben (1997) which proposed 14 relations, *mountain cloud* is classified under the relation of "*head* LOCATED *modifier*", but it does not entail the exact spatial relation between mountain and cloud, which is what speakers naturally would do. The other way of interpretation is free generation of paraphrases, where human annotators paraphrase compounds in their own language. Although this keeps the most semantic information, there can be tons of similar paraphrases for one compound – *golf courses* can be explained as "courses for golf", "courses for playing golf", "courses for the game of golf" (Rambelli et al., 2024). Too much similar information is not ideal for NLP tasks.

To remedy this dilemma, Rambelli et al. (2024) used the Hatcher-Bourque classification method proposed in Pepper (2022) that is said to be cross-linguistically consistent. It lists 29 relations, including 17 relations and the reverse correspondents for some of them, and assigns a paraphrase template to each relation. For example, the relation of USAGE should always be paraphrased as "(a) head that (a) modifier uses" (e.g., *lamp oil*, "the oil that a lamp uses"), while USG-R, the reverse correspondent of USAGE, should always be paraphrased as "(a) head that uses (a) modifier" (e.g., *oil lamp*, "a lamp that uses oil"). This method ensures that each compound only has one paraphrase that captures enough semantic information, and as a result, all compounds with the same relation were paraphrased similarly.

Rambelli et al. (2024) looked through previous datasets and collected lexicalized English noun-noun compounds that have clear mapping between the original relations and the Hatcher-Bourque relations. This resulted in a dataset containing 668 lexicalized noun-noun compounds (henthforth LNC dataset), including 9 out of 29 relations in Hatcher-Bourque classification[1]. Each compound only has one correct relation and one correct paraphrase. Table 1 below includes the count of compounds under each relation, an example compound for each relation, and the corresponding paraphrase.

[1] Although there are 2 more different datasets used in Rambelli et al. (2024), this capstone project is not going to use them, because they are either not focusing on lexicalized compounds, or not using Hatcher-Bourque classification system.

Implicit relations	Count	Example Compound	Paraphrase
COMP(OSITION)-R	85	Plastic bag	A bag that is composed of plastic
CONT(AINMENT)-R	54	Trash bag	A bag that contains trash
LOCATION	107	Supermarket shelf	A shelf that is located in a supermarket
PARTONOMY	16	Car door	A door that is part of a car
PRODUCTION	13	Food company	A company that produces food
PROD(UCTION)-R	47	Bank loan	A loan that a bank produces
PURPOSE	270	Research group	A group intended for research
TOPIC-R	66	Art class	A class that is about art
US(A)G(E)-R	10	Wind turbine	A turbine that uses wind

Table 1: a list of implicit relations found in Rambelli et al. (2024)'s LNC dataset, the number of compounds under each relation, example noun-noun compound for each relation, and its paraphrase

To test whether models can distinguish between the correct and incorrect paraphrases using either surprisal scores or prompt choices as the evaluation, the authors also generated 8 incorrect paraphrases for each compound using the template. In this way, models were always given 9 paraphrases to generate scores or to decide among.

With the design of the dataset introduced, Section 3 will give more details on the surprisal scores so there is a clearer picture on Rambelli et al.'s experiment design which was replicated in this capstone project.

3. Surprisal Scores

As described above, for implicit relations in noun-noun compounds, label prediction is not always an ideal evaluation method. A better solution should focus on the paraphrases that capture more semantic information and test whether models can understand the paraphrases, which requires a method that can look into models' behaviors. The `minicon` library from Misra (2022) solves this problem. It allows researchers to go inside transformer models and conduct behavioral analysis on the models, and `scorer` module in this library includes a list of functions that can extract the word- and sentence-level log probabilities for different types of language models. Surprisal scores are generated from the `scorer` module, and it is defined on the word level (S_t) as the negative of the log probability of a token t_i conditioned on the preceding sentence tokens $t_{<i}$. For a whole sentence, the surprisal scores (S_s) were calculated by the sum of the surprisals of each token (S_t) normalized by the length of the sentence:

$$S_S = \frac{\sum_{t \in S}^{T} S_t}{count(t)}$$

The hypothesis is that for sentences that makes more sense to the model, their surprisal scores are lower than sentences that make less sense. For example, models would have lower surprisal score for "they are coming" than for "they is coming" or "they are came." In Rambelli et al. (2024), for each compound, they extracted the surprisal score of all 9 paraphrases (1 correct and

8 incorrect) and measured whether the correct paraphrases have lower scores. Accuracy was then measured by the number of compounds whose correct paraphrase received the lowest surprisal scores divided by the total number of compounds. In both experiments below, I will use this type of accuracy to evaluate models' performances.

With the background of the LNC dataset and the surprisal score accuracy measurement, the two sections below will give details on the experiments done for this capstone project.

4. Experiment 1: Attention-free Models' Interpretation of English Noun-Noun Compounds

Transformer models get all the attention, but recently some scholars decided that attention is something we don't need. Gu and Dao (2024) proposed a new architecture of Mamba, which is an attention-free selective state space model, and claimed that it can achieve the same performance with the best transformer language models. If that is the case, then on the task of predicting implicit relations in lexicalized English noun-noun compounds, does it perform as well as the transformer models, such as Mistral and Falcon?

To answer this question, I extracted surprisal scores from and prompted on several mamba and transformer models. The Mamba models I used were `falcon-mamba-7b`[2] and its instruction variant `falcon-mamba-7b-instruct` (Zuo et al., 2024)[3], and I compared them with `falcon-7b`[4], `falcon-7b-instruct` (Almazrouei et al., 2023)[5], `mistral-7b-v0.3`[6], and `mistral-7b-instruct-v0.3` (Jiang et al., 2023)[7].

First I used the evaluation of surprisal scores. `scorer` module includes different classes for extracting scores from masked language models (e.g., BERT, `MaskedLMScorer`), incremental language models (e.g., Mistral, `IncrementalLMScorer`), and allegedly mamba language models (`MambaScorer`). However, I failed to run functions in `MambaScorer` after tried all available solutions I could find, so I used functions in `IncrementalLMScorer` to extract mamba models' surprisal scores. There are 2 reasons for this decision: 1) `IncrementalLMScorer` is designed for autoregressive and incremental models and mamba is an autogressive model according to Gu and Dao (2024); 2) I looked through the code and found that `IncrementalLMScorer` uses `transformer.AutoModelForCausalLM` class methods to get the pretrained models and tokenizers and the official HuggingFace documentation for `falcon-mamba-7b` also uses the same class methods.

Surprisal score accuracy for each model is given in Table 2. Among these models, the best accuracy scores came from `falcon-mamba-7b-instruct`, which is 0.4401. Although `falcon-mamba-7b` did not perform the best, it performs as well as other transformer models by 0.4177. This result indicates that on the task of implicit relations prediction in lexicalized

[2] https://huggingface.co/tiiuae/falcon-mamba-7b
[3] https://huggingface.co/tiiuae/falcon-mamba-7b-instruct
[4] https://huggingface.co/tiiuae/falcon-7b
[5] https://huggingface.co/tiiuae/falcon-7b-instruct
[6] https://huggingface.co/mistralai/Mistral-7B-v0.3
[7] https://huggingface.co/mistralai/Mistral-7B-Instruct-v0.3

English noun-noun compounds, mamba models indeed can perform as well as, or even better than, transformer LLMs.

Model	**Accuracy**
falcon-7b	0.4237
falcon-7b-instruct	0.3743
falcon-mamba-7b	0.4177
falcon-mamba-7b-instruct	**0.4401**
Mistral-7B-v0.3	0.4087
Mistral-7B-instruct-v0.3	0.4266

Table 2: Accuracy scores of each model under the evaluation method of surprisal scores

The second evaluation method was to prompt the instruct variants to choose the correct paraphrase. Rambelli et al. (2024) reported that they achieved best performance under 1-shot situation, compared with 3-shot and zero-shot. They also gave an example of the prompt they used, which was integrated into my prompt. A template of my prompt is available in the Appendix. it always starts with Rambelli et al.'s example of *olive oil* and some instructions on what to do to make a decition, followed by the actual question with the target compound. The instruction were given to avoid the model generating answers freely, which is difficult for data analysis later on. The order of the 9 paraphrases (1 correct, 8 incorrect) was randomly generated by running the experiment so that the correct paraphrase is not always under the same option number.

Rambelli et al. (2024) only ran prompts on the instruction variants of the LLMs, so I only prompted the three instruct variants. I could not find Rambelli et al.'s code for prompting LLMs, so I decided to follow the official HuggingFace instructions on text generation for each of them, to achieve better results. For `Mistral-7B-instruct` and `Falcon-7b-instruct` where I can use HuggingFace text generation pipelines, I set the same parameter values[8], but I did not change the parameters when prompting `falcon-mamba-7b-instruct`, because I could not run HuggingFace text generation pipelines on it.

The resulting accuracy of each model is listed in Table 3. Similar to what was reported in Rambelli et al. (2024), `falcon-7b-instruct` has bad performance on this task, with a low accuracy of 0.1287. `falcon-mamba-7b-instruct` performed significantly better than `falcon-7b-instruct` with the accuracy of 0.2979, although compared with `Mistral-7b-instruct-v0.3`'s 0.57 it is not a high score at all. This can only indicate that mamba models perform better than other models that were pretrained with similar datasets, but we cannot generalize that into mamba models perform as well as or even better than transformer models. Unfortunately, companies that own popular LLMs such as Mistral have not yet released mamba models designed for text generation, so we cannot know how well other mamba language models can do on the task of implicit relation prediction.

[8] Temperature: 0, do_sample: True, top-k: 10, top-p: 5, max-tokens: 100.

Model	Accuracy
Falcon-7b-instruct	0.1287
falcon-mamba-7b-instruct	0.2979
Mistral-7B-Instruct-v0.3	0.5749

Table 2: Accuracy scores of each model under the evaluation method of numbers of correct choices

While this section covers the experiment with only English data, Section 5 below covers another experiment that was done on both English and Chinese dataset.

5. Experiment 2: Bilingual or Multilingual BERT vs. Monolingual BERT

Although monolingual English models are the main language spoken by LLMs, some researchers have invented some multilingual transformer models that is said to be fluent in a list of languages and achieve the performance of monolingual models on some tasks. This leads to a similar research question to the last one: on the task of predicting implicit relation in lexicalized noun-noun compounds, do bilingual or multilingual models achieve the same accuracy scores with monolingual models? Besides, do they perform well in both languages, or do they only do well in one?

To answer this question, datasets in two different languages are needed. I decided to continue using the LNC dataset from Rambelli et al. (2024) as in Experiment 1 for time efficiency. After looked through some datasets in other languages, I decided to annotate a small corpus of Chinese lexicalized noun-noun compounds on my own. There is no dataset using the Hatcher-Bourque classification system in previous studies, but because I decided to keep the LNC dataset, I need a dataset with the same set of relations and paraphrases to be able to compare the results. This means if I use a dataset in another language, I need to reannotate them using the Hatcher-Bourque relations, and design paraphrase templates that are natural enough to catch the sematic meaning of each compound. Since I am only fluent enough in Mandarin Chinese to be able to do these tasks, I decided to annotate some Mandarin Chinese compounds and build a dataset.

All Chinese noun-noun compounds come from Ji and Gagné (2007) which is a psycholinguistics research that investigated implicit relations in Chinese compounds. They selected 450 two-character compounds in Modern Chinese Words Dictionary and listed each compound's frequency scores on Google web page. Because LNC dataset contains only lexicalized compounds, I tried to filter out those with extremely low frequency scores of below 10,000 (e.g., *铅人*, "lead person", frequency score 780). Many compounds that share the same character also share the same relation, which may affect models' prediction when it sees a certain character, so I eliminated those as well. Among the remaining ones, compounds with clear verbs were also eliminated (e.g., *习题*, "study questions", frequency score 4,590,000). I also removed some compounds that I was not sure how to interpret, either because they are too lexicalized in Chinese to be interpreted character by character or because they are too rarely seen. Finally, compounds which should be assigned a relation that is not among the 9 relations in the LNC dataset were also removed. As a result, 174 compounds were kept from Ji and Gagné (2007). I later added 2 compounds that were not in the study to provide more data points for the relation of PRODUCTION. Therefore, in this self-created dataset, there are 176 Chinese noun-noun compounds labeled with Hatcher-Bourque classification method.

In terms of paraphrases, I translated the original English template in Pepper (2022) into Chinese, but with some variations. One paraphrase template can fit almost all compounds under one relation in English, but in Chinese, sometimes more than one template is needed for the same relation, or it would sound unnatural. For example, for the relation of PROD(UCTION)-R whose English template is "(a) head that produces (a) modifier", it must have at least 2 Chinese templates to capture the animacy of the modifier. Table 3 shows all 9 relations in the dataset, the count of compounds under each relation, their English templates, and their corresponding Chinese templates.

Implicit relations	**Count**	**English Template**	**Chinese Template**	**Distinction**
COMP(OSITION)-R	31	(a) head composed of (a) modifier	Modifier 做成的 head	Can be decomposed into a bunch of same pieces (e.g., flower bundle)
			Modifier 组成的 head	Cannot be decomposed (e.g., silk shirt)
CONT(AINMENT)-R	14	(a) head that contains (a) modifier	装 modifier 的 head	Physically contains
			有 modifier 的 head	Ingredient contains
LOCATION	27	(a) head located at/near/in (a) modifier	在 modifier 上的 head	Located on
			在 modifier 边的 head	Located near
			在 modifier 里的 head	Located in
			在 modifier 中的 head	Located in (only used for the compound whose modifier means "air")
PARTONOMY	26	(a) head that is part of (a) modifier	Modifier 的 head	
PRODUCTION	4	(a) head that produces (a) modifier	生产 modifier 的 head	
PROD(UCTION)-R	20	(a) head that (a) modifier produces	Modifier 产生的 head	Inanimate
			Modifier 发出的 head	Animate
PURPOSE	34	(a) head that is intended for (a) modifier	为 modifier 设计的 head	The head is not an organization or a job role
			为 modifier 设立的 head	The head is an organization or a job role

TOPIC-R	12	(a) head that is about (a) modifier	关于 modifier 的 head	
US(A)G(E)-R	8	(a) head that uses (a) modifier	用 modifier 的 head	

Table 3: a list of 9 relations in the self-created Chinese dataset, official English paraphrase template in Pepper (2021), their corresponding Chinese paraphrase template(s) I created, and the distinctions

The problem then comes to the number of incorrect paraphrases. In LNC dataset, each compound always has 1 correct paraphrase and 8 incorrect paraphrases, because each relation only has 1 paraphrase template. In my dataset, although each compound also has only 1 correct paraphrase, the number of incorrect paraphrases can be different. If a compound is assigned a relation that has more than one template, the other template(s) will not be included into the incorrect list, or it can be hard to calculate the accuracy scores. Therefore, the number of incorrect paraphrases for each compound can range from 12 to 15.

With both datasets ready, I decided to use the English-Chinese bilingual BERT model `bert-base-en-zh-cased` (Abdaoui et al., 2020). Researchers for this model pretrained a list of smaller multilingual BERT models and claimed that they can achieve similar results with Google's multilingual BERT. In fact, Rambelli et al. (2024) used BERT as the baseline, and it indeed performed awfully (accuracy score 0.262), but I could not find other free-to-use multilingual transformer models fluent in Chinese, and my main goal was to compare multilingual models with monolingual models instead of achieving the best performance, so unideal models can do this, too.

Surprisal scores were then extracted on several BERT models and are listed in Table 4. For English, monolingual BERT model `bert-base-uncased` was chosen to compete with the bilingual model. Although both models did not give ideal performance, their accuracy scores under this evaluation method were similar – bilingual model achieved 0.2874 and monolingual model achieved 0.3174. However, surprisal scores on the Chinese dataset tells a different story. The comparison was made between `bert-base-en-zh-cased` and the monolingual `bert-base-chinese`. Although monolingual model achieved 0.2841, bilingual model only got 0.1023, which is significantly lower. In fact, it is even lower than chance – monolingual English model achieved 0.1705 on Chinese data, and monolingual Chinese model achieved 0.1347 on English data.

Model type	Model name	English (LNC dataset)	Chinese (self-created dataset)
Bilingual	`bert-base-en-zh-cased`	0.2874	0.1023
Monolingual English	`bert-base-uncased`	**0.3174**	0.1705
Monolingual Chinese	`bert-base-chinese`	0.1347	**0.2841**
Multilingual	`bert-base-multilingual-uncased`	0.3099	0.2102

Table 4: Accuracy of monolingual/bilingual/multilingual models on both English and Chinese datasets

The bilingual model was not made by Google and has less parameters, so I tested on Google's multilingual BERT. I put both datasets in and extracted the surprisal scores respectively. Accuracy for English data, 0.3099, remains the same with results from monolingual and bilingual BERT models, but accuracy for Chinese data, 0.2102, is still significantly lower than monolingual Chinese model and only slightly above chance. To sum up the comparisons among all models, bilingual or multilingual models do not always perform on the same level for all languages on the prediction of implicit relations in Chinese noun-noun compounds – they may perform similarly with monolingual models in English, but this conclusion cannot be generalized to other languages, especially not Chinese.

Why did models perform so awfully on the Chinese data? After reading some documentations and papers of the surprisal score extractor (i.e., `minocons` library), I think the most likely reason is that this evaluation method is not designed for Chinese. The extractor for masked language models used the methods proposed by Kauf and Ivanova (2023) which mentioned that their methods are not of much relevance to isolating languages, but Mandarin Chinese is a typical isolating language, so the surprisal scores extracted from BERT for Chinese data may not be able to well-represent the ability of the models. Another evidence of surprisal scores not being the best metric for Chinese is that monolingual Chinese LLMs performed similarly to monolingual Chinese BERT. Rambelli et al. (2024) reported generally better accuracy scores of LLMs compared with BERT in English, so I ran the Chinese data on `Chinese-Mistral-7B-Instruct-v0.2`[9], a Chinese Mistral model. Its surprisal score accuracy was 0.2898, only slightly higher than the monolingual Chinese BERT's 0.2841 – the expected outperformance of LLM disappeared under the evaluation method of the distribution of surprisal scores. It would be nice to see how these models do under prompt evaluation as described in the last section, but it requires multilingual LLMs which I did not manage to find because most reliable ones are not free to use.

[9] https://huggingface.co/itpossible/Chinese-Mistral-7B-Instruct-v0.2

Another potential reason for the overall bad performance of the models may have to do with the dataset. First, it is a small dataset, so it may not be powerful enough to show models' abilities. In fact, looking back at my criteria, I realized that there is no need to exclude compounds that share the same characters and the same relations, because extracting surprisal scores do not require looking through the dataset and thus there is no possibility of models accidentally relating a certain character with a certain relation. However, it was this wrong criterion that filtered out at least 225 out of 450 compounds in Ji and Gagné (2007). Second, although I tried to follow the criteria I listed above when picking compounds, it was not until when I was writing this report that I realized I accidentally missed some unideal ones. For example, I wanted to exclude compounds with frequency scores below 10,000, but at least 2 compounds have scores below that line, which means that there are at least 2 compounds that are unlexicalized. According to Rambelli et al. (2024), models generally perform worse on unlexicalized compounds. They also reported that for novel compounds, BERT performed slightly better than Mistral, which is similar to the lack of difference between these two models described in the last paragraph. Finally, looking back at the annotated data now, I believe there are some correct paraphrases not natural enough to be assigned a lower surprisal score, which makes this dataset even further from being perfect.

The potential explanations for the unideal results give some future directions for this project. First, it is necessary to add more Chinese compounds to the dataset and ask more than one native speaker to translate the template so the paraphrases sounds natural. Those 450 compounds in Ji and Gagné (2007) is still a good starting point. Second, it is necessary to find a multilingual LLM fluent in Chinese, prompt them to choose as in Experiment 1, and evaluate the accuracy. This will say more about models' abilities on interpreting implicit relations in Chinese noun-noun compounds. Finally, if models still perform awfully after these two remedies, then it is necessary to look at other evaluation methods specifically designed for Chinese that can evaluate models' linguistic competency.

6. Conclusion

The main goal of this capstone project is to test different model's ability on interpreting the implicit relations between the head and the modifier in lexicalized noun-noun compounds. Specifically, this project tested the performances of an attention-free mamba model and some bilingual or multilingual transformer models and see if they perform as well as the trending transformer English-only models. To do this, I used the same dataset (LNC dataset) and the evaluation methods (surprisal scores and prompt choices) in Rambelli et al. (2024) which investigated LLMs' performances on English compounds.

First experiment focuses on mamba models, specifically `falcon-mamba-7b` and its instruct variant, intended to answer whether this attention-free architecture can also reach the performance of trending transformer LLMs with the same number of parameters. Both evaluation methods were used. Surprisal scores accuracy indicated that mamba models can perform as well as transformer models and even outperform them, while prompt choice accuracy shows that mamba models can only reach the performance of transformer models that were trained with similar datasets but not all transformers.

Second experiment focuses on bilingual and multilingual BERT models' performance in different languages, intended to test whether they perform the same with monolingual BERTs in all languages. Mandarin Chinese was chosen to be the other language to compare with English. LNC dataset was kept, and a self-created dataset was used for Chinese. All Chinese compounds are from a previous published study and annotated using the Hatcher-Bourque classification method. In terms of evaluations, only surprisal scores were used, because BERT models are not available for prompting and promptable multilingual LLMs are all expensive to use. Results indicate that all models, no matter monolingual, bilingual, or multilingual, performed similarly on the English data. However, for Chinese data, monolingual BERT performed significantly better than other two models, and the bilingual model even performed lower than chance. This shows that on the task of implicit relation prediction, multilingual models do not perform as well as monolinguals for all languages, and this especially the case for Chinese. Potential explanations for this underperformance were given, which inspired some future directions on improving transformer models' performance in Chinese data.

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Appendix: Prompt template in Experiment 1

```
Which is the most likely description of "olive oil"?
1. an oil that uses olives;
2. an oil that is part of olives;
3. an oil that olives produce;
4. an oil that produces olives;
5. an oil that contains olives;
6. an oil that is about olives;
7. an oil that is composed of olives;
8. an oil that is located in olives;
9. an oil intended for olives
If you think an option is the most likely one, then output the
exact number of the option, a period, the exact text of the
option, and a semi-colon.
For example, if you think option 1 '1. an oil that uses olives'
is the best option for the question above, then output 'Answer:
1. an oil that uses olives;'
Which is the most likely description of '(the target compound)'?
1. (paraphrase 1);
2. (paraphrase 2);
3. (paraphrase 3);
4. (paraphrase 4);
5. (paraphrase 5);
6. (paraphrase 6);
7. (paraphrase 7);
8. (paraphrase 8);
9. (paraphrase 9);
Answer:
```